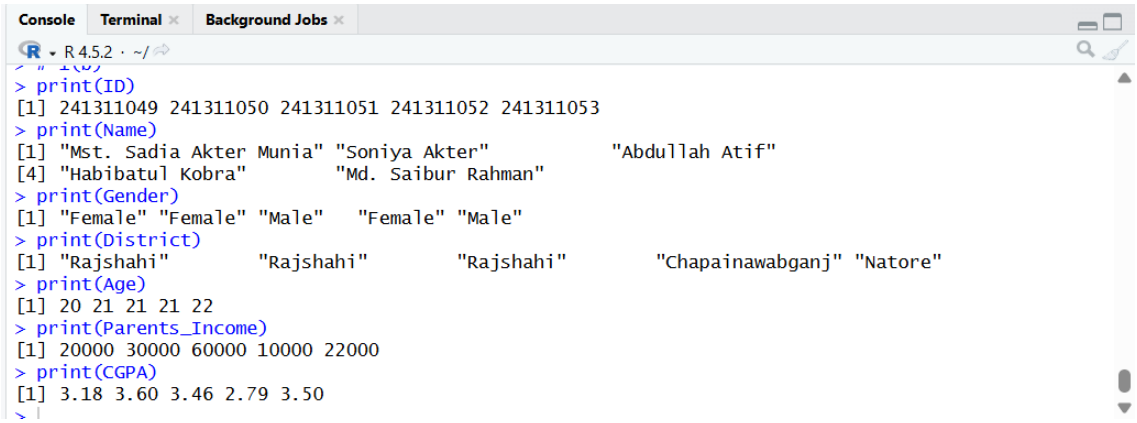
1(a)

```
ID <- c(241311049, 241311050, 241311051, 241311052, 241311053)

Name <- c("Mst. Sadia Akter Munia",

        "Soniya Akter",

        "Abdullah Atif",

        "Habibatul Kobra",

        "Md. Saibur Rahman")

Gender <- c("Female","Female","Male","Female","Male")

District <- c("Rajshahi","Rajshahi","Rajshahi","Chapainawabganj","Natore")

Age <- c(20,21,21,21,22)

Parents_Income <- c(20000,30000,60000,10000,22000)

CGPA <- c(3.18,3.6,3.46,2.79,3.5)
```

1(b)

```
print(ID)

print(Name)

print(Gender)

print(District)

print(Age)

print(Parents_Income)

print(CGPA)
```



1(c)

students <- data.frame(ID, Name, Gender, District, Age, Parents_Income, CGPA)

print(students)

```
> students <- data.frame(ID, Name, Gender, District, Age, Parents_Income, CGPA)
>
> print(students)
         ID                   Name Gender        District Age Parents_Income CGPA
1 241311049 Mst. Sadia Akter Munia Female        Rajshahi  20          20000 3.18
2 241311050           Soniya Akter Female        Rajshahi  21          30000 3.60
3 241311051          Abdullah Atif   Male        Rajshahi  21          60000 3.46
4 241311052        Habibatul Kobra Female Chapainawabganj  21          10000 2.79
5 241311053      Md. Saibur Rahman   Male          Natore  22          22000 3.50
>
```

1(d)

install.packages("psych")

library(psych)

#Arithmetic Mean

mean(Age)

mean(Parents_Income)

mean(CGPA)

```
> #Arithmetic Mean
> mean(Age)
[1] 21
> mean(Parents_Income)
[1] 28400
> mean(CGPA)
[1] 3.306
>
```

#Geometric Mean

geometric.mean(Age)

geometric.mean(Parents_Income)

geometric.mean(CGPA)

```
> #Geometric Mean
> geometric.mean(Age)
[1] 20.99047
> geometric.mean(Parents_Income)
[1] 23974.25
> geometric.mean(CGPA)
[1] 3.292271
>
```

#Harmonic Mean

harmonic.mean(Age)

harmonic.mean(Parents_Income)

harmonic.mean(CGPA)

```
> #Harmonic Mean
> harmonic.mean(Age)
[1] 20.98093
> harmonic.mean(Parents_Income)
[1] 20370.37
> harmonic.mean(CGPA)
[1] 3.277834
>
```

#Median

median(Age)

median(Parents_Income)

median(CGPA)

```
> #Median
> median(Age)
[1] 21
> median(Parents_Income)
[1] 22000
> median(CGPA)
[1] 3.46
>
```

#Mode Function

```
Mode <- function(x){

  ux <- unique(x)

  ux[which.max(tabulate(match(x, ux)))]

}
```

Mode(Age)

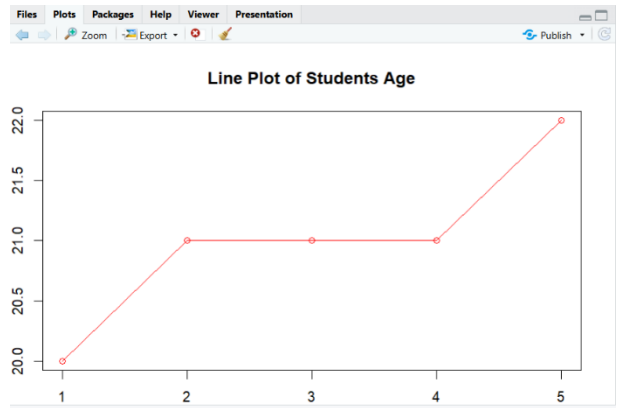
Mode(Parents_Income)

Mode(CGPA)

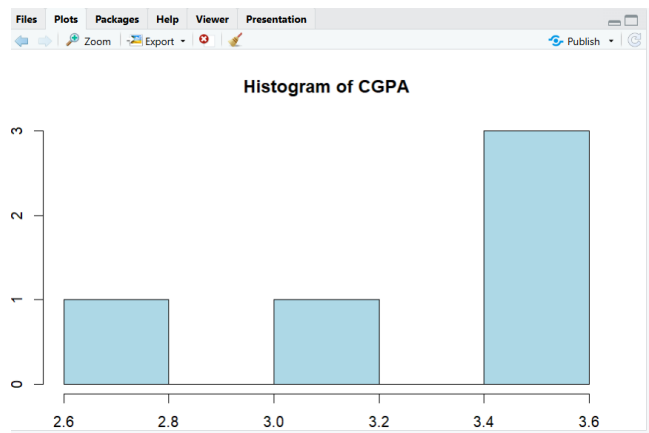
```
> Mode(Age)
[1] 21
> Mode(Parents_Income)
[1] 20000
> Mode(CGPA)
[1] 3.18
>
```

1(e)

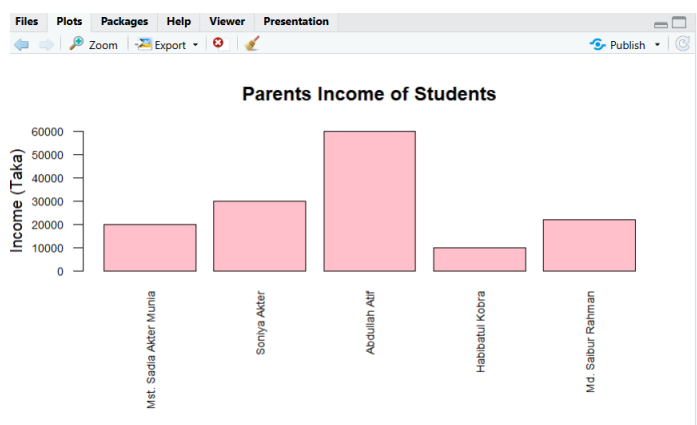
```
#Line Plot
plot(Age,
    type="o",
    col="red",
    main="Line Plot of Students Age",
    xlab="Students",
    ylab="Age")
```



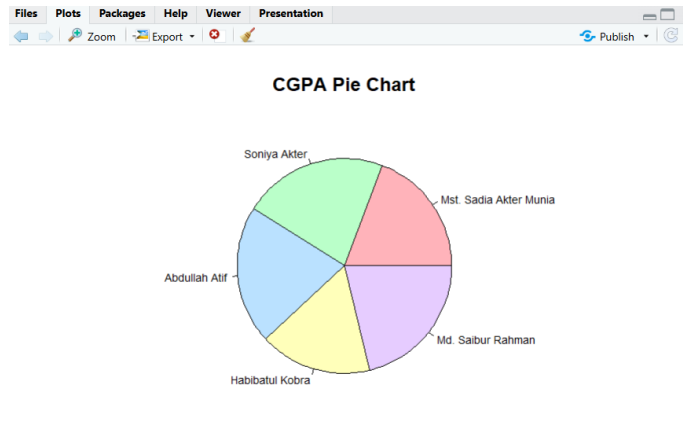

```
#Histogram
hist(CGPA,
    main="Histogram of CGPA",
    col="lightblue",
    xlab="CGPA")
```




```
#Bar Plot
par(mar=c(10,4,4,2))
barplot(Parents_Income,
        main="Parents Income of Students",
        col="pink",
        names.arg=Name,
        las=2,
        cex.names=0.7,
        cex.axis=0.7,
        ylab="Income (Taka)")
```



```
#Pie Chart
par(mar=c(2,2,4,2))
colors <- c("#FFB3BA", "#BAFFC9",
"#BAE1FF", "#FFFFBA", "#E6CCFF")

pie(CGPA,
   labels=Name,
   main="CGPA Pie Chart",
   col=colors,
   cex=0.7)
```



2(a)

```
X <- c(
  11,17,15,22,11,16,19,24,29,25,
  18,18,25,26,32,14,17,20,23,27,
  30,12,24,36,18,15,20,28,33,38,
  34,13,12,16,20,22,29,19,23,31
)
X
```

```
> X
 [1] 11 17 15 22 11 16 19 24 29 25 18 18 25 26 32 14 17 20 23 27 30 12 24 36 18 15 20 28 33 38
[31] 34 13 12 16 20 22 29 19 23 31
>
```

2(b)

```
#Arithmetic Mean
mean(X)
```

```
> mean(X)
[1] 22.05
>
```

#Geometric Mean

library(psych)

geometric.mean(X)

```
> library(psych)
> geometric.mean(X)
[1] 20.88281
>
```

#Harmonic Mean

harmonic.mean(X)

```
> harmonic.mean(X)
[1] 19.7358
>
```

#Median

median(X)

```
> median(X)
[1] 21
>
```

#Mode

```
Mode <- function(x){

  ux <- unique(x)

  ux[which.max(tabulate(match(x, ux)))]

}
```

Mode(X)

```
> Mode(X)
[1] 18
>
```

2(c)

breaks <- seq(min(X), max(X), by=5)

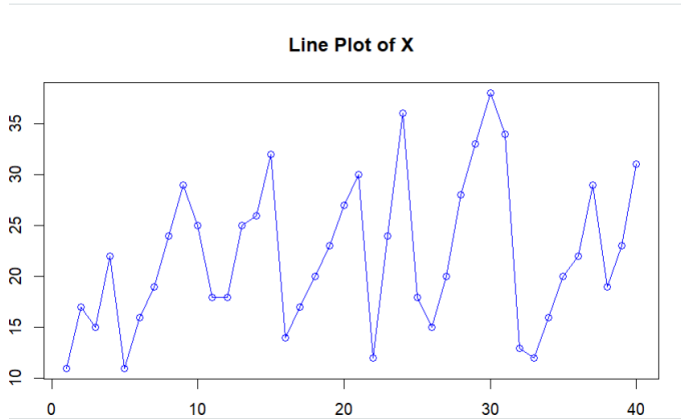
freq_table <- table(cut(X, breaks))

freq_table

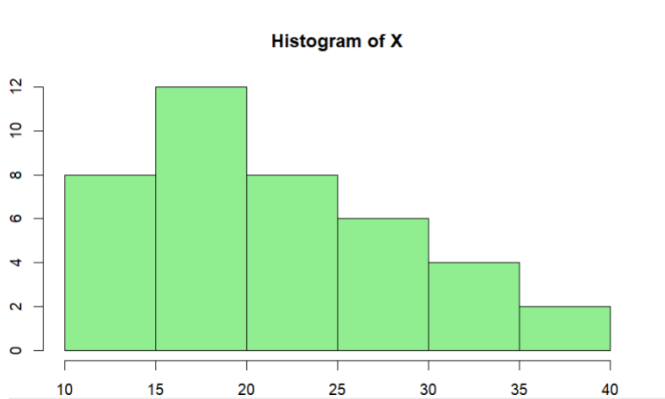
```
> freq_table

(11,16] (16,21] (21,26] (26,31] (31,36]
      8      10       9       6       4
>
```

2(d)

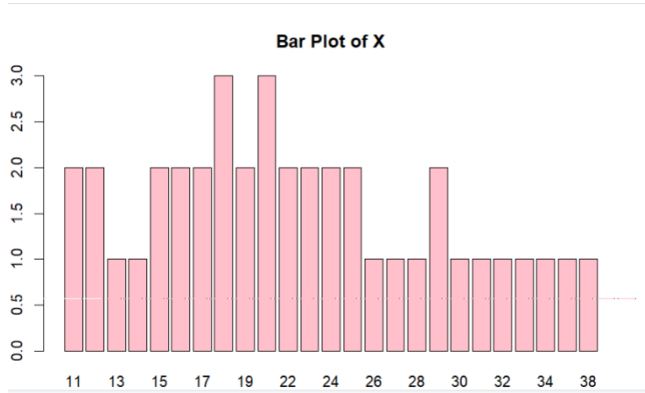
```
#Line Plot
plot(X,
    type="o",
    col="blue",
    main="Line Plot of X",
    xlab="Observation",
    ylab="Value")
```



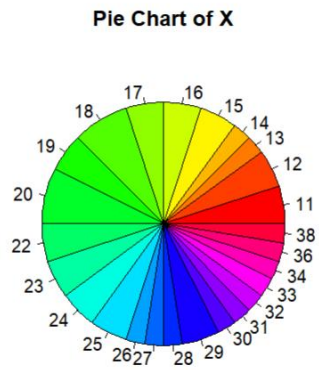

```
#Histogram
hist(X,
    col="lightgreen",
    main="Histogram of X",
    xlab="Values")
```




```
#Bar Plot
barplot(table(X),
        col="pink",
        main="Bar Plot of X",
        xlab="Values",
        ylab="Frequency")
```

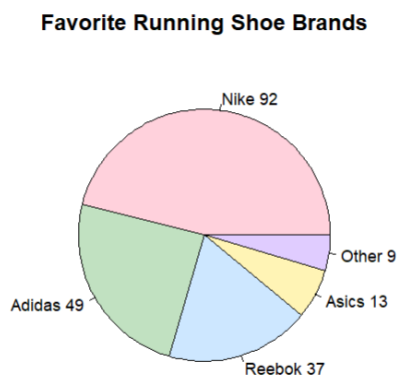


```
#Pie Chart
pie(table(X),
   main="Pie Chart of X",
   col=rainbow(length(unique(X))))
```



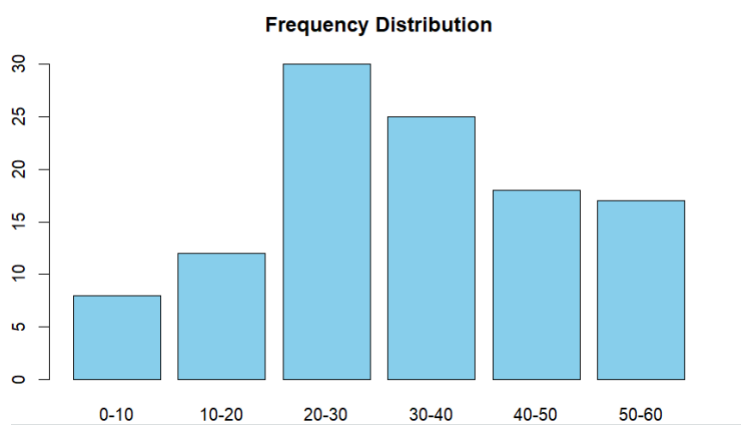
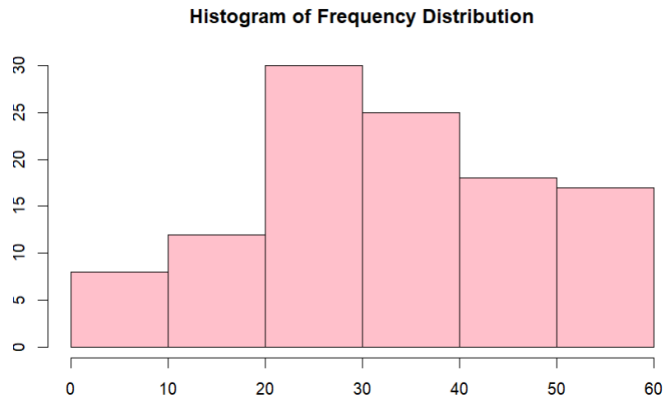
3(a)

```
shoes <- c(92, 49, 37, 13, 9)
brands <- c("Nike", "Adidas", "Reebok", "Asics", "Other")
par(mar=c(2,2,4,2))
colors <- c("#FFD1DC","#C1E1C1",
"#CDE7FF","#FFF4B5","#E0CCFF")
pie(shoes,
   labels=paste(brands, shoes),
   main="Favorite Running Shoe Brands",
   col=colors,
   cex=0.9)
```



3(b)

```
class <- c("0-10","10-20","20-30","30-40","40-50","50-60")
frequency <- c(8,12,30,25,18,17)
#Bar plot
barplot(frequency,
      names.arg=class,
      col="skyblue",
      main="Frequency Distribution",
      xlab="Class Interval",
      ylab="Frequency")
```



```
#Histogram
data <- rep(c(5,15,25,35,45,55), frequency)
hist(data,
    breaks=6,
    col="pink",
    main="Histogram of Frequency Distribution",
    xlab="Class Interval")
```